

CHE 210

1. Course number and name

CHE 210. Material and Energy Balances

2. Credits and contact hours

Credit hours: 4; Contact hours: 150 min/wk. lecture, 50 min/wk. discussion; Engineering

3. Instructor's or course coordinator's name

Lewis Wedgewood

4. Textbook, title, author, publisher, year

Introduction to Chemical Processes: Principles, Analysis, Synthesis by Regina M Murphy, McGraw-Hill, 1st edition, New York, 2007

5. Specific course information

a. Brief description of the content of the course (catalog description)

Material and energy balances applied to chemical systems. Introduction to chemical and physical properties. Introduction to the use of computers for chemical process calculations.

b. Prerequisites or co-requisites

Credit or concurrent registration in CHE 201 and CHE 205.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

- Synthesize a block flow diagram from a process description
- Identify and modify reaction pathways to achieve specific production while minimizing waste and occurrence of hazardous compounds
- Perform a degree of freedom analysis to determine degree of process specification
- Perform component, element, and total mass (or molar) material balances
- Quantify and make appropriate use of system performance specifications for reactors, splitters, separators
- Correctly identify modes of operation for process units (steady-state, transient, batch, continuous) and simplify the general material balance equation appropriately.
- Perform unit conversions as necessary.
- Calculate reaction equilibrium constants and determine fractional conversion and selectivity in equilibrium reactions.
- Use model equations and graphical data to determine composition in vapor-liquid phase equilibria
- Formulate process analysis problems as a system of linear equations
- Use tabulated data or model equations to evaluate changes in state properties
- Perform energy balances to determine basic heating/cooling requirements of a reactive process

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X	X	
(3) an ability to communicate effectively with a range of audiences.	X			
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	X			
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	X			
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	X			

7. Brief list of topics to be covered

- Raw materials, reaction path synthesis
- Generation-consumption analysis
- Material balance equations
- Process flow sheeting
- Degree of freedom analysis
- Reactor materials balance
- Stream composition and system performance (conversion, selectivity, yield)
- Recycle, purge
- Chemical reaction equilibrium and chemical kinetics
- Separation technologies
- Phase equilibria
- Energy balances, Energy flows
- Energy data and model equations
- Energy balances on reactive processes